

FINAL CIRCUITS

SUPPLY INTAKE POINT

Position

Supply authority wants it in the most convenient place in the building, so that the cost of the supply equipment e.g. cable is kept to a minimum.

Engineer on behalf of client wants a suitable position so that electrical installation can be planned with the greatest economy.

Meter reader wants it accessible.

In urban areas underground system is opted (armoured cable). The junction which the cable makes with the street-main is contained in a tee-box, buried outside the premises.

In rural areas overhead system is opted. Service cables are connected to the line conductors by means of mechanical connectors called line-taps. Conductors are always insulated and taken to insulators on the roof/wall of the premises and then run to supply intake position. There must be provision for effective control and protection.

The IEE requires that;

- i. Every Consumer's installation must be adequately controlled by specified equipment
- ii. The equipment must be readily accessible to the consumer
- iii. The equipment should be as near as possible to the supply-intake cut-outs.

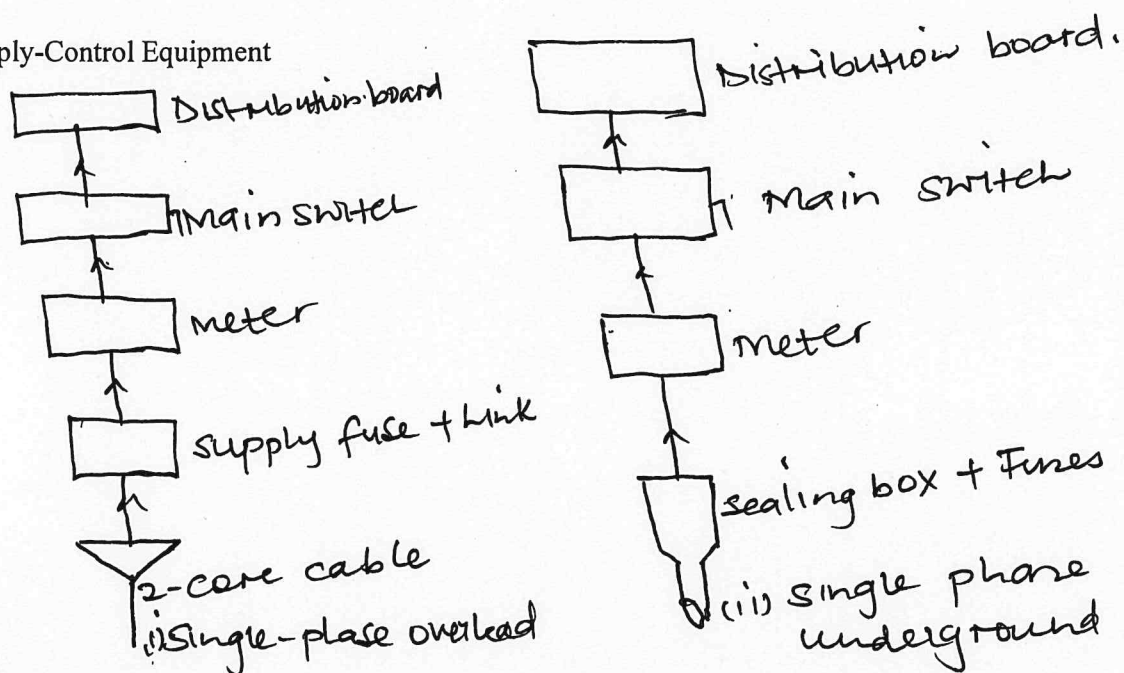
Electricity regulation of the Factories Act says that efficient means (switchgear), suitably located, shall be provided for cutting all pressure from every part of a system, as may be necessary to prevent danger.

Switchgear is a combination of electrical disconnect switches, fuses or circuit breakers used to control, protect and isolate electrical equipment.

The main switchgear in any installation must be able to:

- a) Isolate the complete installation from the supply
- b) Protect the installation against excess current
- c) Cut off the current should a serious earth fault occur

Sequence of Supply-Control Equipment



1. Incoming cable

If overhead line is used, there is a large main fuse cutout and generally a neutral connector-block.

If underground, termination is in the form of a sealing-box combined with a cutout unit. Sealing box serves to seal the end of the cable to prevent moisture from entering into the cable.

2. Cut-out equipment (service fuse and neutral link).

Fuse (60A Or 80A).

Neutral link is a solid flat tinned copper, used where the neutral side of the supply is effectively earthed, otherwise two fuses, one in each conductor are provided

3. Meter

Records the amount of electrical energy used by the consumer.

4. Main Switch or switch fuse

From the meter, the cable (main) goes to main switch which should be double-pole, linked blade type to isolate the complete installation from the supply when the switch is operated.

Single phase –two poles

Three- phase-TPN (triple pole with neutral). All three or four poles are broken

Fuse in the consumer's main switch should be of the same rating as the service fuse (in the live conductor)

Switch fuse (switch and fuse combined in one unit)

5. Distribution board (DFB -Distribution fuse board) or CCU (consumer control unit)

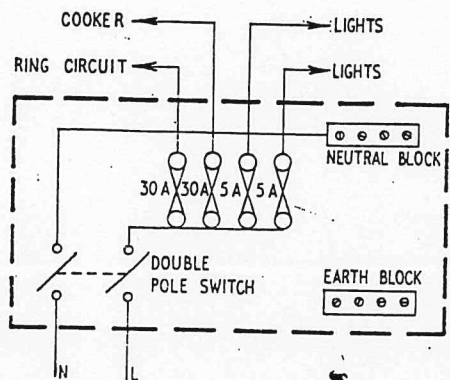
Distributes the electricity to the various circuits making up the complete installation. The circuits are provided with individual fuses, placed in the 'live' conductors, the neutral takes the form of a connector bar or block.

A 'splitter' unit is a combined main switch and FB. They are suitable for small lighting and heating installations.

Disadvantage of a splitter unit: Necessary to switch off all circuits before the lid can be opened to replace a fuse.

An MCB- Distribution board may also be used. It contains MCBS (miniature circuit breakers) instead of fuses which gives overcurrent protection.

A CCU consists of a main switch that isolates both conductors



1. Domestic consumer's control unit.

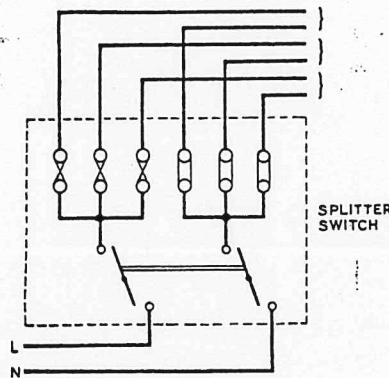


Fig. 2.14. Circuit of splitter switch

Main cables are those which carry the total current of the installation from the cutouts and meters to the main switch through to the distribution board.

FINAL CIRCUIT

A final circuit is one that is connected to its own way in the distribution board, and is not split up to supply or feed other circuits (different circuits have different current rating, easy control and troubleshooting.)

The IEE requires that:

- a. Where an installation comprises more than one final circuit, each shall be connected to a separate way in a distribution board.
- b. The wiring of each final circuit shall be electrically separate from that of every other final circuit. The neutral conductors shall be connected at the distribution board in the same order as that in which the live conductors are connected to the circuit breakers.
- c. The current rating of a fuse for a final circuit shall not exceed the current rating of the lowest-rated cable in the circuit protected by that fuse.

GROUPS OF FINAL CIRCUITS

1. Rating not exceeding 15A
2. Rating exceeding 15A
3. Rated for 13A fused plugs
4. Rated for feeding fluorescent and other discharge-lamp circuits.
5. Rated for feeding a motor.

Rating not exceeding 15A

Examples:	current demand assumed
15A socket outlet	15A
13A Socket outlet	13A
5A Socket outlet	5 A
Lighting outlet	Equivalent to the connected load with a minimum of 100w per lamp holder
2A socket —outlet	At least 0.5A

The IEE requires that:

- i. The number of points which may be supplied is limited by their aggregate demand as the table above.
- ii. There must be no allowance for diversity factor
- iii. Current rating of the cable must not be exceeded.

It is recommended that lightning circuits should be at least two so that should one circuit fail, the other circuit would allow repairs to be carried out.

Lighting circuits(domestic), rated 5A (1.00mm² or 1.5mm² cable)

Lighting circuits (Industrial Application), rated at 15A (2.5mm² or 4.0mm² cable)

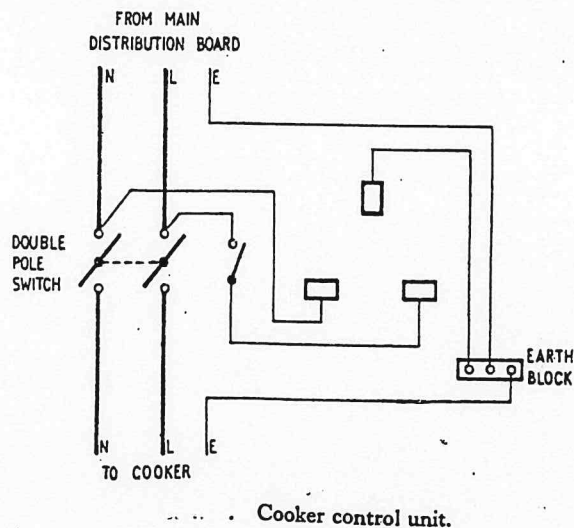
Electrical clock and shavers

Rating exceeding 15A

Cookers----- 30A

Water heater-----20/30A

IEE requires that a final circuit rated at over 15A must not feed more than one point except where there is socket-outlet integral with a cooker unit, or circuits feeding 13A socket outlets, used with fused plugs.



Rated for 13A fused plugs

13A fused plugs use socket-outlets

Plug contain a cartridge fuse

Socket outlets may be connected in ring circuit or radial circuit

Radial: Each sockets outlet is supplied through the previous one and the circuit is supplied from one direction only

Ring: Similar to radial except that the final socket outlet is wired back to the supply position that is any outlet is supplied from two directions

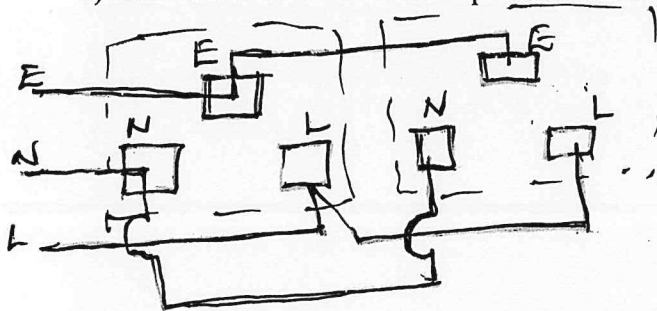
Table A.3M

description	Minimum copper size pvc or rubber cables	Fuse rating	Maximum socket outlets and stationary appliances of rating not exceeding 13A
Radial final circuit, floor area less than 30 m ² not kitchen	2.5mm ²	20A	6 provided that no fixed water heating appliances shall be connected to any of these points.
Radial circuit serving other room (not above)	2.5 mm ² 4.0 mm ²	20 30	2 6
Ring circuit with spurs if any	2.5mm ²	30	A 38-42

SPUR: A branch cable connected to a ring circuit.

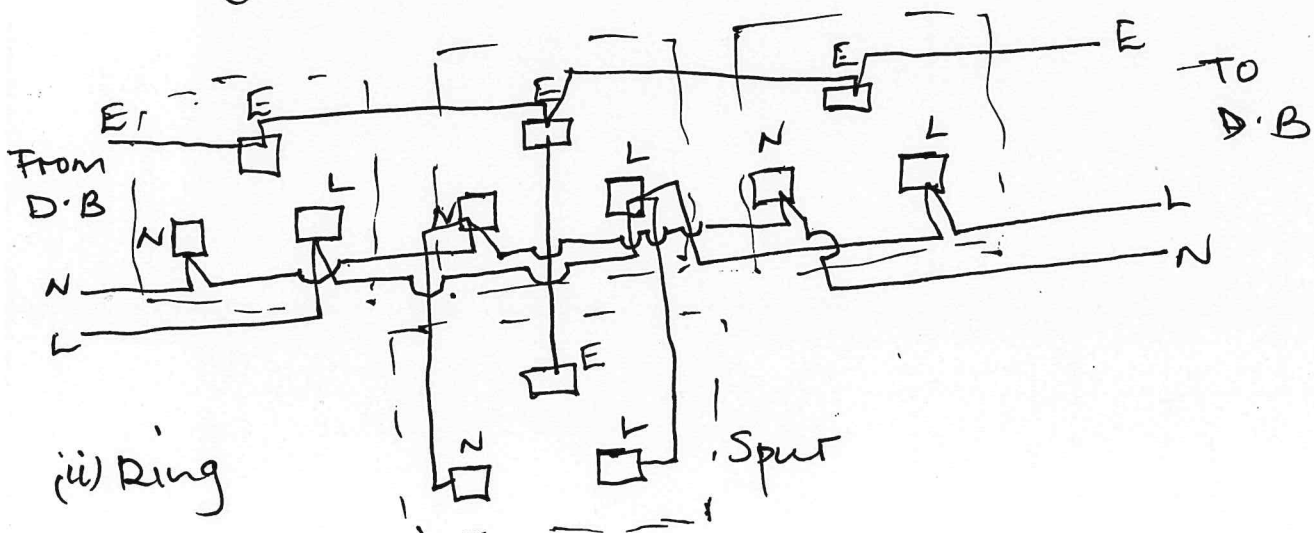
IEE requires that:

- For a ring circuit, the total numbers of spurs shall not exceed the total no of socket outlets and stationary appliances connected directly in the ring.
- Each socket of a twin or multiple socket-outlet unit shall be counted as one socket-outlet.



(i) Radial

D.B → Distribution board.



(ii) Ring

Rated for feeding fluorescent and other discharge lamp circuits

IEE requires that;

- i. Circuits shall be capable of carrying the total steady current i.e. that of the lamps and any associated gear. Lamp watts multiplied by not less than 1.8. i.e. a circuit is corrected to a power factor of not less than 0.85. and takes into account control gear losses.
- ii. Any switch controlling circuits comprising discharge lamps should have a rating of not less than twice the steady current in the circuit it controls, unless the switch is specially designed to break an inductive load at its full rated capacity.

Rated for feeding a motor

IEE Requires that;

- i. Every electric motor shall be provided with means for starting and stopping, so placed as to be easily operated by the person in charge of the motor.
- ii. Every motor shall be provided with control apparatus i.e.
 - a. Means to prevent automatic restarting after a stoppage due to a drop in voltage or failure of supply, where unexpected restarting of the motor might cause danger.
 - b. Where more than one method of manually stopping a motor is provided and danger might be caused by the unexpected restarting of the motor, means to prevent restarting until every stopping device has been reset.
 - c. Means of isolation suitably placed and connected that all voltage may thereby be cut off from the motor and all apparatus including any automatic circuit breaker, used therewith
- (iii) Every electric motor having a rating exceeding 0.37KW shall be provided with control apparatus incorporating a suitable device affording protection against excess current in the motor or in the cables between the device and the motor.

More than one motor may be connected to 15A final circuit provided that the total full-load rating of the motors does not exceed the rating of the smallest cable in the circuit.

If the rating of the motor exceeds 15A, then the circuit must supply one motor only.